Dr. K. Rajmohan

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PROFILE

Associate Professor at PERI Institute of Technology, Chennai, with over 11 years of teaching experience in electronics and communication engineering. My research focuses on wavelet-based image segmentation, medical and satellite image processing, signal processing, and VLSI design. I use MATLAB to develop and evaluate algorithms for filtering, feature extraction, noise reduction, edge detection, and pattern recognition.

RESEARCH INTERESTS

Wavelet-based image segmentation, Medical image processing, Satellite image processing, Signal processing, VLSI design

RESEARCH METRICS

Citations: 0 | 0
Publications: 14 (Journals 6, Conferences 7, Book chapters 1)
Research output: Patents: 2 | 0 | 0

EDUCATION

Ph.D. in Electronics and Instrumentation Engineering 2018
Annamalai University
Thesis: Investigations on Wavelet based Image Segmentation techniques

M.E. / M.Tech in VLSI Design 2011
Kamban Engineering College, Tiruvannamalai | 7.3 CGPA

B.E. / B.Tech in Electronics and Communication Engineering 2008
Kamban Engineering College, Tiruvannamalai | 70.2%

PROFESSIONAL EXPERIENCE

Associate Professor 03 October 2024 - Present
PERI Institute of Technology, Chennai
- Teaching and research in Electronics and Communication Engineering; research interests include signal and image processing and VLSI design.

Assistant HR Manager 01 January 2022 - 01 October 2024
Arunai Medical College and Hospital, Tiruvannamalai

Assistant Professor 04 July 2011 - 24 December 2021
Arunai Engineering College, Tiruvannamalai

PATENTS

- "Energy Harvesting Communication modules for IoT devices" - 19 November 2024
- "Computer-Implemented System for Analyzing Human Brain Images using Functional Imaging Data" - 27 March 2020

PUBLICATIONS

Journal Papers (6)

J1. Ravi Kishore M, Shankar K, **Rajmohan K**, Pavan Kumar Reddy Y, and Sunanda Y, "Robust Fusion of Multimodality Medical Images with Fidelity Driven Optimization and Detail Preservation," Springer Nature Link, 2025.

J2. Pavan Kumar Reddy Y, Shankar K, Sunanda Y, Ravi Kishore M, and **Rajmohan K**, "IoT Based Smart Home using Alexa," Springer Nature Link, 2025.

J3. **Rajmohan K**, Thirugnanam G, and Mangaiyarkarasi P, "Hybrid Segmentation Technique based on Dual Tree Complex Wavelet Transform and Watershed Algorithm for Satellite Images," Journal of Chemical and Pharmaceutical Sciences, Vol. 10(2), pp. 253-260, 2017.

J4. **Rajmohan K**, Thirugnanam G, and Mangaiyarkarasi P, "Medical Image based Segmentation using DTCWT and Marker Controlled Watershed Algorithm," Journal of Chemical and Pharmaceutical Sciences, Vol. 10(3), pp. 327-334, 2017.

J5. **Rajmohan K**, Thirugnanam G, and Mangaiyarkarasi P, "Region based Segmentation for Medical Images," International Journal of Exploring Emerging Trends in Engineering, Vol. 3(4), pp. 304-309, 2016.

J6. **Rajmohan K**, Thirugnanam G, and Mangaiyarkarasi P, "Wavelet Packet and Watershed Transform Based Hybrid Image Segmentation For Satellite Image," World Applied Sciences Journal, Vol. 34(8), pp. 1139-1143, 2016.

Conference Papers (7)

C1. Shankar K, **Rajmohan K**, Gnanamurugan S, Fayaz Begum, Maruthi Prasad, and Pavan Kumar Reddy Y, "IoT based delivery Robot with theft detection for food and goods supply," in E3S Web of Conferences, 2025.

C2. Gnanamurugan S, Shankar K, **Rajmohan K**, Mathubala R S, Hemalatha R, and Mariyammal A, "Design and Compact of Microstrip Patch Antenna RT/ DUROID 5880 using FPGA," in International Conference on Advances in Electrical and Computer Technologies, July 24-25, Tiruchengode, 2025.

C3. Rajat Kumar Dwibedi, V Senthil Kumar, **K Rajmohan**, A Durga Bhavani, S Murugesan, Mohit Tiwari, "Hybrid Optimization Algorithms for Dynamic and Static Power Reduction in Low-Power VLSI Circuits," in International Conference on Advances in Computing, Communication and Materials (ICACCM), Nov 2024, 2024.

C4. **Rajmohan K**, "Extraction of features from images based on Adaptive regularized level set method," in National Conference on Cutting Edge Technologies in Electrical and Electronics Systems, March 2, 2018, Cuddalore, 2018.

C5. **Rajmohan K**, Thirugnanam G and Mangaiyarkarasi P, "Hybrid Segmentation technique using wavelet packet and watershed transform for medical images," in International Conference on Advances in Computing and Data Sciences, Nov 11-12, 2016, Ghaziabad, 2016.

C6. **Rajmohan K**, "Distributed arithmetic based low power and low area adaptive FIR filter implementation in FPGA," in National Conference on Recent Advances in Communication Engineering, March 6, 2014, Arni, 2014.

C7. **Rajmohan K** and Thirugnanam G, "A Dualistic Sub-Image Histogram Equalization based Enhancement and Segmentation Techniques for Medical Images," in International Conference on Image Information Processing, Dec 9-11, 2014, Shimla, 2013.

Book Chapters (1)

BC1. **Rajmohan K**, Thirugnanam G, and Mangaiyarkarasi P, "Hybrid Segmentation technique using Wavelet Packet and Watershed transform for medical images," in Communication in Computer and Information Sciences, Vol. 1, pp. 132-139, 2017.

ADDITIONAL INFORMATION

Tools & software: MATLAB, Digital filtering, Fourier analysis, Wavelet analysis, Feature extraction, Noise reduction, Edge detection, Pattern recognition, Time-frequency analysis, Real-time data visualization

Languages: English, Tamil